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Restraint effects in portal frame bridges

Practical design methodologies and background information

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Restraint effects in design of portal frame bridges

Practical design methodologies and background information

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Mario Plos

Report TVBK-3073

LUND UNIVERSITY

LTH FACULTY OF ENGINEERING

CHALMERS

Restraint effects in design of portal frame bridges

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Foreword

This report presents methodology, results and conclusions from the PhD-project “Evaluation, modeling and handling of restraint stresses in concrete bridges”, which can be relevant for bridge designers. The project investigated thermal actions in portal frame bridges, and the widths of cracks resulting from restraint effects. The main results and conclusions are given in the summary in Swedish, while the following parts in English presents both results and conclusions, as well as the methodology.

The report is developed as a guide for practicing bridge designers on how the load case describing temperature differences between structural parts can be included in design. It also highlights the difficulties and need for further research on determining what the resulting crack widths are and how to design crack distribution reinforcement when cracking is caused by restraint effects.

The research was performed at Lund University and Chalmers University of Technology during the period 2014-2019. A more extensive presentation of the theory and research methodology used, as well as the results, can be found in the dissertation resulting from the PhD-project, entitled “Restraint Effects in Concrete Bridges” [1]. The dissertation is in turn based on four journal articles [2-5] and two conference articles [6-7]. Two other conference articles [8-9] were also produced within the project. However, these two articles do not present new information in relation to the journal articles.

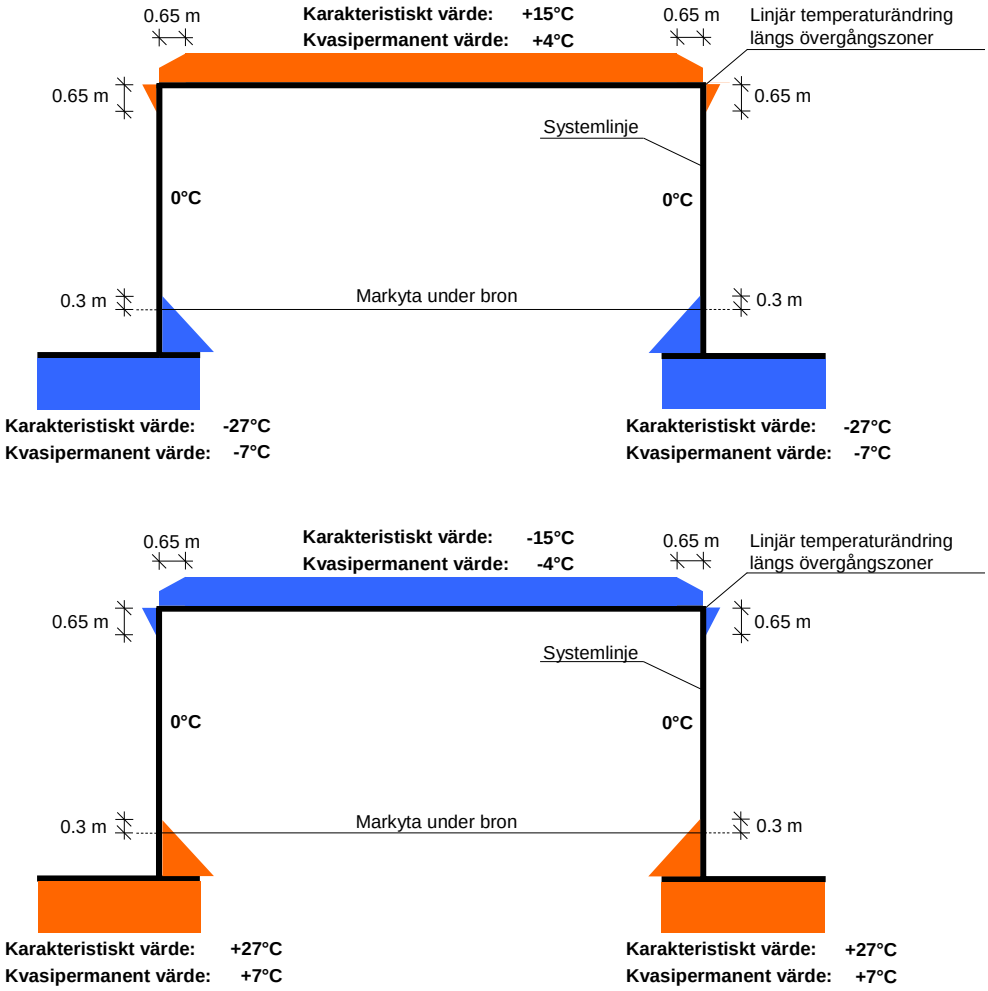
This work was financed by Trafikverket (BBT) and SBUF, whose support is much appreciated. A special thanks to the supervisors of the PhD-project not listed as authors, Miklós Molnár, Annika Mårtensson and Roberto Crocetti, and the reference group, Hans-Ola Öhrström (Trafikverket), Fredrik Carlsson (Trafikverket), Karl Lundstedt (Skanska), Morgan Johansson (Norconsult), Sven Thelandersson (LTH), Johan Kölfors (Scanscot).

Sammanfattning

Denna rapport sammanfattar kortfattat de resultat som erhölls i forskningsprojektet “Utvärdering, hantering och modellering av tvångskrafter i betongbroar”. Inom projektet togs ett förslag fram för hantering av det temperaturlastfall som beskriver skillnader i temperatur mellan olika konstruktionsdelar (avsnitt 6.1.6 i Eurokod 1-5 [10]) för plattrambroar i Sverige. Även sprickvidden då tvångseffekter orsakar uppsprickning undersöktes, och det visades att sprickvidden är svår att uppskatta utan att ta hänsyn till den spänningsminskning som sker vid uppsprickning. Hur detta kan tas hänsyn till vid dimensionering av plattrambroar är emellertid inte till fullo utrett på grund av problemets komplexitet och svårigheter att koppla spänningsminskningen till enskilda faktorer.

Temperaturlastfall

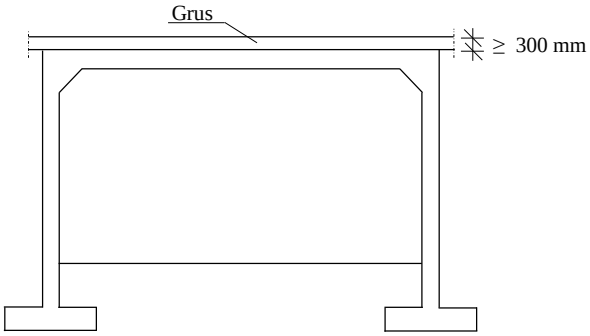
Det föreslagna temperaturlastfallet visas i figuren på nästa sida. Det beskriver två olika situationer: en situation när brobaneplattan är den varmaste konstruktionsdelen samtidigt som bottenplattan är kallast, och en situation när situationen är den omvända. I ramhörnen och i de nedre delarna av rambenen så är övergångszoner markerade, längs med vilka temperaturen varierar linjärt. Temperaturen antas vara konstant över tvärsnittshöjden på alla konstruktionsdelar i detta lastfall, då gradienter är beskrivna i andra temperaturlastfall i Eurokod 1-5 [10], som inte motsvarar avsnitt 6.1.6. Både karakteristiska och kvasipermanenta lastvärden ges. De karakteristiska är tänkta till att för de flesta fall användas för att kontrollera om uppsprickning sker över huvud taget, och de kvasipermanenta är huvudsakligen ämnade för sprickviddsberäkning. De angivna temperaturlastvärdena beskriver den relativa temperaturen, dvs skillnaden i temperatur mellan konstruktionsdelarna, och anger därmed inte temperaturen i absoluta tal. Temperaturlastfallet behöver i normala fall inte tas med i beräkningar i brottgränstillståndet, eftersom tvångsspänningarna minskar kraftigt när stora deformationer uppstår och när omfattande uppsprickning sker. Dessutom är de rörelser som konstruktionen eftersträvar på grund av lastfallet så pass små att de inte kan leda till brott i konstruktionen.



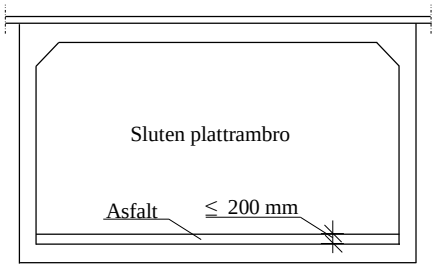
Lastfallet är ett allmänt lastfall som kan användas för de flesta plattrambroar i Sverige med traditionell utformning. Utöver detta lastfall har två specialfall studerats, för vilka temperaturskillnaderna kan minskas. Det gäller broar med minst 300 mm grus på brobaneplattan, vilket ofta gäller för järnvägsbroar, och slutna plattrambroar (dvs. broar med en gemensam bottenplatta som upplag för båda rambenen) med maximalt 200 mm asfalt på bottenplattan. För det första fallet kan temperaturskillnaden mellan brobaneplatta och ramben minskas, och för det andra fallet kan temperaturskillnaden mellan ramben och bottenplatta minskas. De lastvärden som föreslås för dessa konstruktionstyper visas i tabellerna nedan. För broar som uppfyller båda villkoren kan båda speciallastfallen användas samtidigt. Övergångszonerna definieras på samma sätt för specialfallen som för det allmänna lastfallet. Det finns även andra sorters specialfall som inte har undersökts i denna studie, exempelvis snedvinkliga plattrambroar och plattrambroar med välvda ramben. Emellertid förväntas inte temperaturskillnaderna mellan konstruktionsdelarna bli väsentligt annorlunda för dessa fall. Däremot kan tvångssituationen förändras.

Nedan illustreras de geometrier som specialfallen gäller för, och därunder ges lastvärdena i två tabeller, en för fallet då brobaneplattan är varmast och bottenplattan är kallast, och en för det omvända fallet.

Specialfall 1: ≥300 mm grus på brobaneplattan. Minskad temperaturskillnad mellan brobaneplatta och ramben



Specialfall 2: Sluten plattrambro och ≤200 mm asfalt på bottenplattan. Minskad temperaturskillnad mellan ramben och bottenplatta



Temperaturlastvärden för fallet då brobaneplattan är varmast och bottenplattan är kallast:

	Allmänt lastfall	**≥300 mm grus på brobaneplattan**	**Sluten plattrambro med ≤200 mm asfalt på bottenplattan**	**Sluten plattrambro med ≤200 mm asfalt på bottenplattan och ≥300 mm grus på brobaneplattan**
Karakteristisk temperatur, brobaneplatta	+15°C	+10°C	+15°C	+10°C
Kvasipermanent temperatur, brobaneplatta	+4°C	+2°C	+4°C	+2°C
Karakteristisk temperatur, bottenplatta	-27°C	-27°C	-10°C	-10°C
Kvasipermanent temperatur, bottenplatta	-7°C	-7°C	-2°C	-2°C

Temperaturlastvärden för fallet då brobaneplattan är kallast och bottenplattan är varmast:

	Allmänt lastfall	**≥300 mm grus på brobaneplattan**	**Sluten plattrambro med ≤200 mm asfalt på bottenplattan**	**Sluten plattrambro med ≤200 mm asfalt på bottenplattan och ≥300 mm grus på brobaneplattan**
Karakteristisk temperatur, brobaneplatta	-15°C	-10°C	-15°C	-10°C
Kvasipermanent temperatur, brobaneplatta	-4°C	-2°C	-4°C	-2°C
Karakteristisk temperatur, bottenplatta	+27°C	+27°C	+10°C	+10°C
Kvasipermanent temperatur, bottenplatta	+7°C	+7°C	+2°C	+2°C

Tvångssprickor

Temperaturrörelser och krympning ger upphov till tvångskrafter i konstruktioner som är förhindrade från att röra sig. I fallet då olika konstruktionsdelar av en plattrambro har olika temperatur eller har utvecklat olika mycket krympning, exempelvis för att de gjöts vid olika tidpunkter, så kommer konstruktionsdelarna sträva efter att ha olika längd i brons breddriktning. Eftersom konstruktionsdelarna är fast inspända i varandra kommer de att begränsa varandras möjligheter till rörelser, vilket ger upphov till tvångskrafter. Tvångskrafternas storlek beror på konstruktionsdelarnas styvhet, vilket innebär att då uppsprickning sker och styvheten minskar, så minskar även tvångskrafterna. Vid dimensionering baserad på exempelvis FE-modell med linjärelastisk materialmodell och sprickviddsberäkning enligt avsnitt 7.3.4 i Eurokod 2-1-1 [11] tas ingen hänsyn till denna styvhetsminskning. Detta innebär att metodiken överskattar sprickvidden ifall den orsakats av tvångseffekter.

Forskningsstudien som denna rapport baseras på har visat att signifikanta tvångseffekter på grund av skillnader i temperatur och krympning mellan konstruktionsdelar endast uppstår i plattrambroars breddriktning. Det är osannolikt att de undersökta lasterna ger uppsprickning i brobaneplattan, sprickor kan däremot uppstå både i överkant och underkant av rambenen. De största sprickorna väntas uppkomma i rambenens underkant, där dragspänningar kan uppstå av den kombinerade effekten av temperaturlastfallet och en skillnad i krympning på grund av att bottenplattan gjutits före ramben och brobaneplatta. Uppsprickning i bottenplattorna har inte undersökts i studien, men förväntas bli mindre allvarlig än i rambenen, då en eventuell skillnad i krympning ger tryckspänningar i bottenplattorna. Observera att dessa resultat endast är giltiga för situationer där byggnationen gjorts i två gjutetapper, med olika tidpunkter för gjutning av bottenplattor och ramben samt brobaneplatta. Skillnaden i krympning orsakad av ett kortare gjutuppehåll mellan gjutning av ramben och brobaneplatta anses vara försumbar.

De sprickvidder som erhölls med olinjära FE-analyser med de allmänna temperaturlastfallet och skillnad i krympning mellan ramben och bottenplatta blev mindre än motsvarande sprickor när linjärelastisk beräkning kompletterad med sprickviddsberäkningar enligt Eurokod 2-1-1. Det visade sig också att armeringsmängden hade en mindre inverkan på sprickvidden i de olinjära beräkningarna än vad som ges i uttrycken i Eurokod 2-1-1. Därmed är en ökning av armeringsmängden inte ett lika effektivt sätt att minska sprickvidder vid tvångssituationer som då uppsprickning orsakats av yttre laster. Det är istället mer effektivt att reducera tvånget genom exempelvis ändrad konstruktiv utformning.

Vid dimensionering av broar är det av praktiska skäl inte lämpligt att göra den typen av olinjära FE-analyser som användes i forskningsprojektet. Att ta fram en metodik för hur tvångssprickors bredd kan beräknas vid dimensionering visade sig vara komplext då det inte går att på ett enkelt sätt koppla nivån av uppsprickning till enskilda faktorer. Två tänkbara vidare strategier är att (1) justera den sprickvidd som fås med linjärelastisk finita-elementmodell och sprickviddsberäkning enligt Eurokod 2-1-1 med ett matematiskt uttryck, eller att (2) hantera sprickor på grund av tvång i brons tvärriktning i en separat analytisk beräkning. Hur dessa strategier skulle se ut är emellertid en fråga för fortsatt forskning.

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1. Introduction

Thermal actions and shrinkage lead to a desire for movements in structures. If these movements are prevented by e.g. adjacent structures or structural parts or, in the case of shrinkage, by the reinforcement, stresses will form in the structure. Such stresses are called restraint stresses, as they are caused by the structure being restrained from adjusting its shape. In a portal frame bridge consisting of a slab, abutments and foundations, the concrete structural parts are rigidly connected, which means that adjacent parts can impose movements on each other, especially along the rigid connections, i.e. in the transverse direction of the bridge.

The magnitude of restraint stresses depends on the magnitude of the restrained strain, but also on the stiffness of the structure. When reinforced concrete cross sections crack, the stiffness is reduced, which thus in turn reduces the restraint forces. This is beneficial in a design perspective as the crack width will be smaller than it would have been otherwise. However, as the restraint is reduced for every new crack in a non-uniform way, the final number of cracks is very hard to determine beforehand, and thus also the magnitude of the stress reduction. In Sweden, there is no common methodology for considering the effect of cracking on the restraint effects in the design of bridges. Some designers might disregard it completely, which leads to an overestimation of the restraint forces, while others may assume that the stabilized crack pattern is reached, which risks underestimating the restraint forces.

Another uncertainty is the load values for the thermal actions and shrinkage used in design. In this project, a suggestion of a national adaptation of the thermal load case in section 6.1.6 in Eurocode 1-5 [10] is given, for use in design of portal frame bridges. The Eurocode load case describes differences in temperature between structural parts of bridges. It is stated in the load case that a temperature difference of 15°C should be applied between “main structural elements” of bridges. According to Section 2 of the same document, the given 15°C load value can be assumed to be a characteristic value. However, the load case does not describe how the temperature varies spatially within the structural parts. It is also unclear what “main structural elements” correspond to in different bridge types. The load case is exemplifying these elements as “e.g. tie and arch” which is also an unclear formulation. However, as there are no ties and arches in portal frame bridges, there is reason to believe that the load case was developed with other bridge types in mind.

When the Eurocode load case is used in design of portal frame bridges, it can cause large stresses in the transversal direction. This issue is most common in crack width limitation, which is made in the serviceability limit state using quasi-permanent load values. Such results were not obtained when 2D-frame design models were used, as the effect of the load case in the transversal direction was disregarded with such a model. As this change in prescribed reinforcement was an effect of the change in design procedures, it has been questioned whether the increased reinforcement amounts arising from the use of 3D-design models are reasonable.

This document presents a suggestion of a Swedish national choice for the load case 6.1.6 in Eurocode 1-5 [10] for design of portal frame bridges, see Chapter 2. It also gives more background information to the results, which is presented in Chapter 3. In Chapter 4, a summary of the results found on crack widths when cracking is caused by restraint effects is given. These results indicate that the conventional design approach overestimates the required reinforcement when cracking is caused by restraint, but no simple ways to consider this effect while still using linear elastic design models were found.

2. Suggested Swedish adaptation of thermal load case

In this chapter, the suggested Swedish national adaptation of the thermal load case in section 6.1.6 in Eucocode 1-5 [10] is presented. The national adaptation is constituted of one general load case for portal frame bridges presented in Section 2.1, and two special load cases presented in Section 2.2, which can be applied to portal frame bridges with certain geometries. The thermal load cases are based on a traditional geometry for portal frame bridges, with foundations, abutments, wing walls and bridge decks as illustrated in Figure 1.

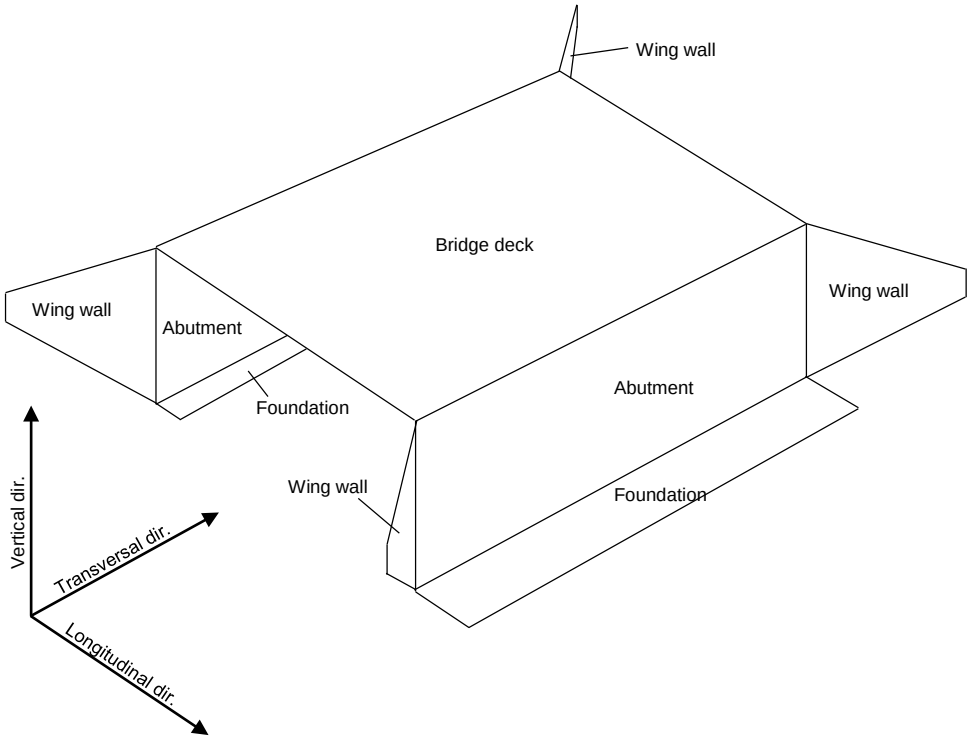


Figure 1. Fundamental geometry of a portal frame bridge, with structural parts marked.

2.1. Thermal actions in Eurocode

Thermal actions are presented in Eurocode 1-5 [10], and for bridges, several different types of thermal load cases are presented in Section 6 of the document. Section 6.1.3 describes the calculation of uniform thermal actions for bridges, which are determined considering the bridge type and location. The calculation of temperature gradients over the bridge deck are presented in Section 6.1.4. These values are calculated considering the type of bridge deck as well as of the type of pavement being placed on the bridge. Either a linear or a bi-linear gradient can be determined. Section 6.1.5 describes how the uniform thermal action and the gradient over the bridge deck can be combined, by reducing one of the two types of actions by multiplication with factors equaling either 0.35 or 0.75.

Section 6.1.6 describes the load case which was investigated within this study, i.e. temperature differences between structural parts of bridges. These thermal effects should be combined with the uniform load case, although since the effects in the transverse direction for the uniform load case is limited, it can be reasonable to assume that the load case presented here have the main influence on the results. Finally, section 6.2 of Eurocode 1-5 considers gradients over the cross-sectional height of bridge piers. To the author's knowledge, this load case is often not considered in design of portal frame bridges in Sweden.

The thermal load case describing differences in temperature between structural parts must be included in the serviceability limit state (SLS). However, it normally does not need to be included in the ultimate

limit state (ULS). The reason for this is that it causes a desire for movements, but that these movements can be expected to occur before failure is reached due to concrete cracking and reinforcement yielding. Consequently, it does not contribute to cause failure in structures in ULS. Shrinkage differences between structural parts should be included in SLS but can normally be omitted in ULS for the same reason.

2.2. General load case

The general load case is presented in Figure 2, and was developed using the methodology presented in Chapter 3. The general load case describes both the situation where the bridge deck is the warmest structural part and the foundation is the coldest, and the opposite case, when the bridge deck is coldest and the foundation is warmest. The temperature in the abutments and the wing walls is zero in both cases, while one of the two other structural parts shown is warmer while the other is colder. As the load case describes temperature differences, only the relative temperatures of the structural parts are of interest. The temperature in the abutment has therefore been set to 0°C, to make the temperature differences easier to read.

According to the general load case, the temperature changes linearly along the transition zones in the bottom of the abutments and in the corner regions. The length of these transition zones are given in Figure 2. Including these transition zones in design reduces the transversal tensile stresses caused by the load case, and also makes the largest stress values appear at more realistic locations in the structure.

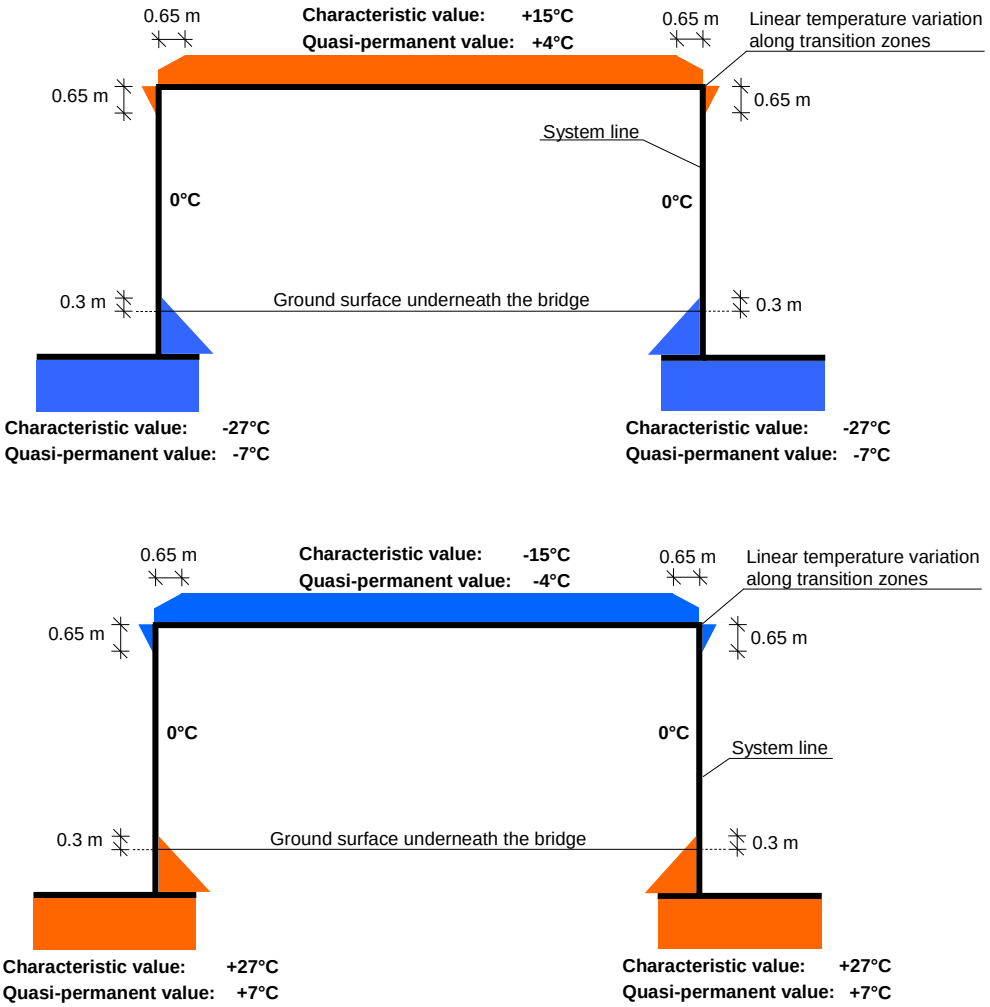


Figure 2. The general thermal load case suggested to be used in design of portal frame bridges. The figure only shows the relative temperature of the parts, and not the temperature in absolute numbers.

Some comments regarding the use of the general load case are given below:

- Both the case when the bridge deck is warm and the foundation is cold, and the case when the bridge deck is cold and the foundation is warm, shall be included in design.
- The characteristic load values can be used to determine whether cracking needs to be accounted for in SLS and, consequently, if crack widths need to be checked at all.
- The quasi-permanent load values are meant to be used in e.g. crack width calculation.
- The general load case defines the structural parts in the cross section as bridge deck, abutment and foundation. The load case is not developed for any other division of structural parts.
- The length of the transition zones in the frame corner is independent of cross-sectional thickness and haunches in the upper corners of the bridge.
- The length of the transition zone in the bottom of the abutment is only dependent on the level of the ground surface below the bridge, and has been shown to be independent of the cross-sectional height of the abutment.
- The general load case is developed for the case with asphalt paving on the bridge deck, but can also be applied for the case with concrete paving or gravel. Using these other paving materials are likely to reduce the temperature difference between bridge deck and abutment due to the lower absorptivity of the materials and/or the larger thicknesses. A special load case with reduced load values was developed for portal frame bridges with at least 300 mm gravel on the bridge deck, see Section 2.2.
- The temperature difference between abutment and foundation given in the general load case can be used regardless of foundation depth, and can be considered a worst case scenario. However, the actual temperature difference between abutment and foundation varies with the depth of the foundation and also with its horizontal position. Accordingly, an extension of the foundation in horizontal direction underneath the bridge deck reduces the temperature differences. A special load case with reduced load values was developed for portal frame bridges with single foundations, see Section 2.2.

2.3. Special load cases

In two special load cases, which can be used for certain portal frame bridges, the load values given in the general load case are reduced. The first special load case regards bridges with at least 300 mm gravel placed on the bridge deck, thus often applicable for railway bridges. For this situation, the temperature difference between bridge deck and abutment can be reduced, see Figure 3a.

The second load case regards portal frame bridges with a single foundation for both supports (in Swedish called “sluten plattrambro”), for which the foundation is only covered by an asphalt paving with a thickness of no more than 200 mm, see Figure 3b.

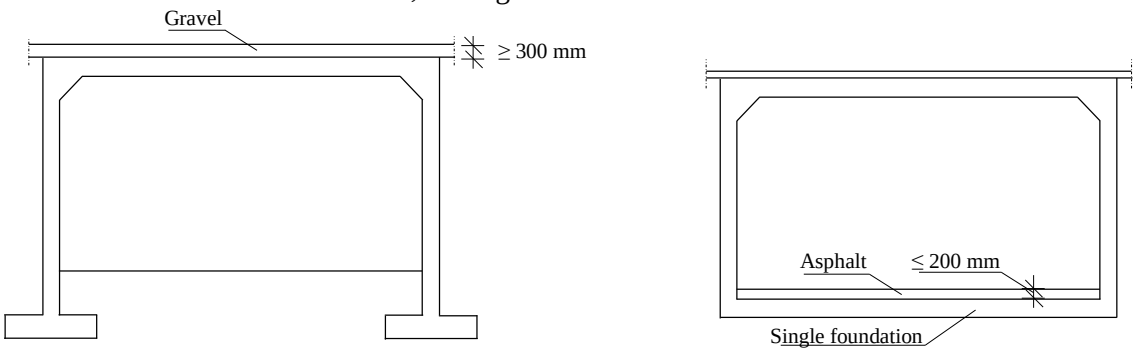


Figure 3. Illustrations of geometries where the special load cases regarding gravel covered bridges (a) and single foundation bridges (b) are applicable.

Compared to the general load case described in section 2.1, only the load values are changed in the special load cases, i.e. the transition zones are defined the same way as in the general load case. The load values are also presented in the same way, with 0°C in the abutment and wing walls, while one of the other structural parts is warmer and the other is colder. Both the case when the bridge deck is the warmest part and the case when the bridge deck is the coldest part should be included in design, just as for the general load case. The cases are presented in Table 1 and 2. For a bridge which fulfills the conditions of both special cases, the reduced load values can be combined.

There are also other types of special cases which have not been investigated in this study, e.g. skewed portal frame bridges and portal frame bridges with cambered abutments. However, the temperature differences between the structural parts are not expected to become significantly different for these cases. The restraint stresses can on the other hand develop differently.

Table 1. Thermal load values for the load case with the bridge deck being the warmest structural part.

	General load case	**≥300 mm gravel on the bridge deck**	**Single foundation with ≤200 mm asphalt on the foundation**	**Single foundation with ≤200 mm asphalt on the foundation and ≥300 mm gravel on the bridge deck**
Characteristic temperature, bridge deck	+15°C	+10°C	+15°C	+10°C
Quasi-permanent temperature, bridge deck	+4°C	+2°C	+4°C	+2°C
Characteristic temperature, foundation	-27°C	-27°C	-10°C	-10°C
Quasi-permanent temperature, foundation	-7°C	-7°C	-2°C	-2°C

Table 2. Thermal load values for the load case with the bridge deck being the coldest structural part.

	General load case	**≥300 mm gravel on the bridge deck**	**Single foundation with ≤200 mm asphalt on the foundation**	**Single foundation with ≤200 mm asphalt on the foundation and ≥300 mm gravel on the bridge deck**
Characteristic temperature, bridge deck	-15°C	-10°C	-15°C	-10°C
Quasi-permanent temperature, bridge deck	-4°C	-2°C	-4°C	-2°C
Characteristic temperature, foundation	+27°C	+27°C	+10°C	+10°C
Quasi-permanent temperature, foundation	+7°C	+7°C	+2°C	+2°C

2.4. Summary

In this chapter, an adaptation of the load case for temperature differences between structural parts (section 6.1.6 in Eurocode [10]) is presented. The load case developed to be used in design of portal frame bridges in Sweden. Both characteristic and quasi-permanent load values are presented, as characteristic values can be used to investigate whether cracking is likely in SLS, and quasi-permanent values are meant to be used in crack width calculations. The load case consists of one general load case, which can be used for portal frame bridges in general, and two special load cases developed for special types of portal frame bridges, for which the load values can be reduced.

3. Background information to the suggested thermal load cases

3.1. Temperature simulations

The thermal load cases presented in Chapter 2 are based on thermal simulations using weather data. A model for simulating temperature in a concrete slab surrounded by air, initially presented by Larsson [12], was used, and extended into a model of a portal frame bridge cross section with adjacent soil and fill [2]. The model was thereafter verified by measuring temperature in 12 locations in a portal frame bridge cross section, and using the analysis model for simulating the temperature during the same period of time. The comparison showed that the analysis model can be used to estimate the difference in temperature between bridge deck and abutment in portal frame bridges, the difference between measured and simulated value being up to 1.5°C at a given point in time, while the average difference over time being close to zero. For this reason, a safety margin of 1.5°C was recommended when determining extreme values (e.g. characteristic values), and no safety margin was recommended when determining time average values (e.g. quasi-permanent values) [3]. The safety margin is thus included in the characteristic load values presented in Tables 1 and 2.

After the model was verified for simulation of temperature in portal frame bridges for Swedish climate, a parametric study was carried out to investigate how various geometrical parameters affected the temperature difference. The parametric study was done using weather data from a time period during summer and another time period during winter. It was found that a relatively small bridge, with thin structural parts was most prone to large temperature differences. For this reason, this type of bridge was used in the continued studies. Further, the portal frame bridge was divided into structural parts and transition zones, as was shown in Figure 2. The temperature differences were defined as the difference in mean temperature between the defined structural parts. The transition zones were thereby not included in the structural parts in this aspect, as that would risk underestimating the temperature difference. Thereafter, eight data series available from SMHI with all necessary weather data (air temperature, wind speed, solar radiation and long-wave radiation) from a time period of at least ten years were used in the simulation model, in order to calculate the hourly temperature difference. The locations and time periods of the data series are shown in Table 3.

Table 3. SMHI weather data series used in thermal simulations.

Location	Data period	No. of years
Tarfala	2008 – 2017	10
Kiruna	2008 – 2017	10
Luleå	1988 – 1997	10
Svenska Högarna	2008 – 2017	10
Stockholm	1986 – 1997	12
Norrköping	2008 – 2017	10
Visby	2008 – 2017	10
Lund	1983 – 1999	16

3.2. General conclusions from temperature simulations

An example of how the temperature difference between the structural parts may look like is given in Figure 4, which shows the temperature difference between bridge deck and abutment, and abutment and foundation, for the entire simulation with data from Norrköping. Positive values indicate that the bridge deck is warmer than the abutment, or that the abutment is warmer than the foundation. Negative values indicate the opposite situations. Several conclusions can be drawn from the figure, such as:

- In summer, the bridge deck is generally warmer than the abutment, and the abutment is warmer than the foundations. In winter, the abutment is instead warmer than the bridge deck, and the foundation is warmer than the abutment.

- During spring and fall, the temperature differences between the structural parts are less extreme, as they are transitioning from one of the situations mentioned above into the other.
- The temperature differences are generally larger between the abutment and foundation than between the abutment and bridge deck, for the geometry used in the analyses.
- The temperature difference fluctuates less in the abutment-foundation case, as the temperature of these parts changes more slowly over time than in the bridge deck.

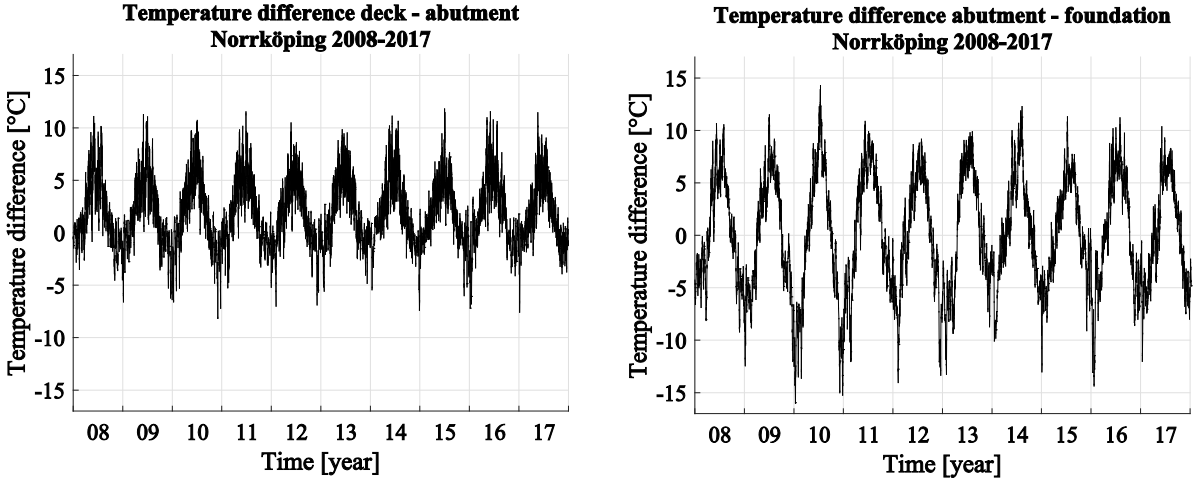


Figure 4. Temperature difference between bridge deck and abutment (left) and abutment and foundation (right) using weather data from Norrköping from the period 2008-2017.

3.3. Determination of load values

Using the results shown in Figure 4 and the corresponding results from the other locations, characteristic and quasi-permanent load values were determined. The fact that positive and negative values in Figure 4 cause tensile stresses in different parts of the bridge (i.e. in the colder part) means that the quasi-permanent load value cannot be determined as just the mean value over time. If the mean or average value was used, as is suggested as the definition of quasi-permanent loads for e.g. live loads in buildings, the obtained load value would either be positive or negative. This would in turn mean that the other situation is completely disregarded. In order not to underestimate the long-term effects of the load, it was instead chosen to divide each temperature difference (deck-abutment and abutment-foundation) into two cases, one for positive values and one for negative. When determining positive quasi-permanent load values, mean values of the positive values during the periods from April to September were used, and when determining negative values, mean values of the negative values during the periods from October to March were used.

The characteristic values determined shall correspond to values with a reference period of 50 years. In order to derive such values from data series with 10 to 16 years of data, the data series were divided into reference periods. The largest (or smallest) value of each reference period, i.e. the extreme values, were listed, and a general extreme value distribution function was thereafter fitted to the extreme values. The load value corresponding to the 50-year load was thereafter determined from the function. 3-day reference periods from half of each year only (the half where the load is dominating, the same periods as used for the quasi-permanent load) were chosen, although other periods were tested. The development of the load values is described in more detail in [4].

The resulting quasi-permanent and characteristic load values for the data series used are shown in Tables 4 and 5, respectively. From these tables, the largest values for each situation have been chosen to constitute the load case. Also, the same value is used for positive and negative temperature differences, for simplicity.

Table 4. Quasi-permanent load values for each data series.

Data series	Quasi-permanent temperature difference, deck – abutment		Quasi-permanent temperature difference, abutment – foundation	
	Positive	Negative	Positive	Negative
	Value [ºC]	Value [ºC]	Value [ºC]	Value [ºC]
Tarfala 2008-2017	3.5	-1.4	5.0	-4.1
Kiruna 2008-2017	3.3	-1.8	6.1	**-5.2**
Luleå 1983-1997	3.4	**-1.8**	**6.5**	-4.6
Svenska Högarna 2008-2017	2.7	-0.8	4.8	-3.8
Stockholm 1983-1997	3.3	-1.3	5.9	-3.8
Norrköping 2008-2017	**4.0**	-1.1	5.1	-4.3
Visby 2008-2017	3.1	-0.8	4.9	-3.9
Lund 1984-1999	3.4	-1.0	5.3	-3.3

Table 5. Characteristic load values for each data series.

Data series	Characteristic temperature difference [°C], deck – abutment		Characteristic temperature difference [°C], abutment – foundation	
	Positive	Negative	Positive	Negative
Tarfala 2008-2017	13.5	-9.7	15.1	-16.7
Kiruna 2008-2017	14.0	**-13.7**	**18.5**	-25.0
Luleå 1988-1997	12.0	-12.0	17.3	**-26.7**
Svenska Högarna 2008-2017	11.4	-6.1	14.6	-15.8
Stockholm 1986-1997	13.0	-10.5	16.8	-21.7
Norrköping 2008-2017	14.1	-11.2	15.4	-21.8
Visby 2008-2017	11.8	-7.5	15.4	-17.5
Lund 1984-1999	**14.7**	-9.5	14.9	-17.5

3.4. Dependency of result on air temperature

As can be seen in Tables 4 and 5, the obtained load values varied between the different locations. This could be considered in a future development of regional load values for the load case. As data from more locations are required in order to develop regional load values, and as all available data series of at least 10 years of length have been used in this study, calculated values of solar-and long wave radiation could instead be used, as investigated in [13]. However, it was also found that the results obtained in this study for the positive and negative temperature difference between abutment and foundation correlate with the difference in mean air temperature between July and January. This is shown for quasi-permanent load values in Figure 5. Correspondingly, the negative temperature difference between bridge deck and abutment correlates with the temperature fluctuations during January, when this load is generally most

active. This is illustrated for the case with quasi-permanent load values in Figure 6, where the negative temperature difference is plotted against the difference between the maximum and minimum temperature during January, calculated as the mean value for the years of data included in the study. However, the positive temperature difference between bridge deck and abutment did however not seem to correlate with the air temperature fluctuations in July, or any other easily calculated temperature variation. This is possibly explained by the solar radiation having a large influence on this result.

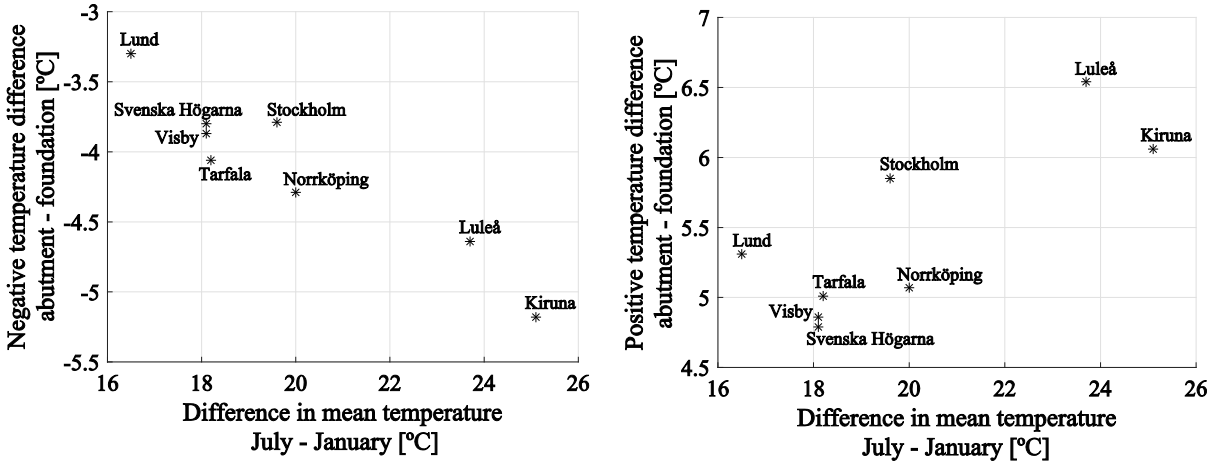


Figure 5. Relation between seasonal temperature variations and the quasi-permanent load value for temperature differences between abutment and foundation.

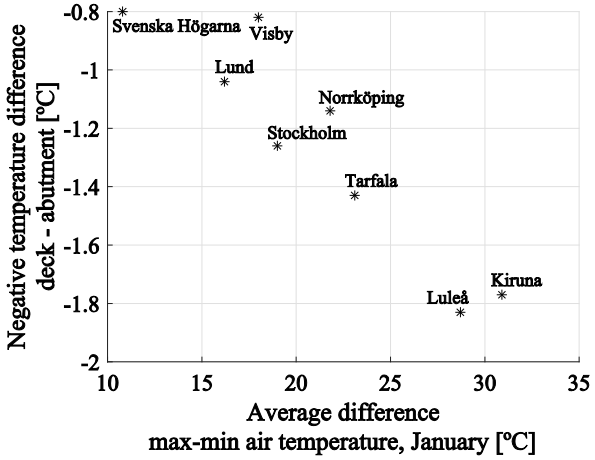


Figure 6. Relation between the characteristic load value for negative temperature difference between bridge deck and abutment, and temperature variations during the month when the load is largest.

3.5. Transition zones

The length of the transition zones in the suggested load case is motivated by analyses of the temperature variation along the system line of the structure. Further, additional thermal simulations were carried out in which the geometry was varied. These analyses were made with shorter periods of weather data, which had been shown to cause large temperature differences in the original structural model. An example of results from these analyses is given in Figure 7, showing results from when the cross-sectional thickness of bridge deck and abutment was varied, for analyses with data series causing large positive and negative temperature differences, respectively. The horizontal axis shows the distance along the system line, from mid-span to the end of the foundation. The figure also illustrates the areas (in dark grey) included in the "structural parts" when calculating their respective mean temperatures. The transition zones used in the design model are between points B and D, and E and G, respectively.

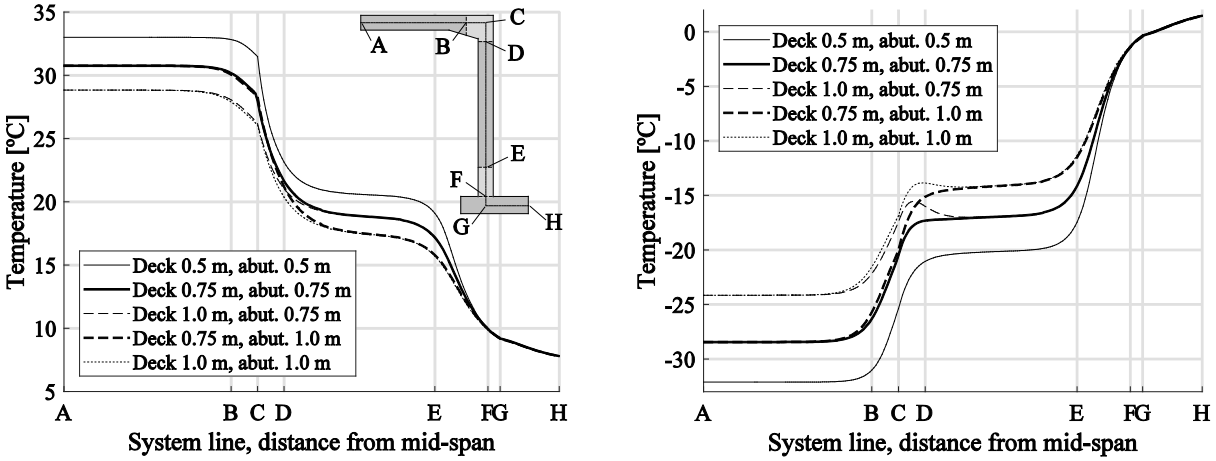


Figure 7. Temperature along the system line when the cross-sectional thicknesses of bridge deck and abutment were varied.

Figure 7 shows that the changes in temperature occur more or less along the same distance, i.e. within the transition zones, regardless of the cross-sectional thickness. There is however a difference in the actual transition length in the corner region depending on which part is warmer. When the bridge deck is warmer, the temperature change occurs mainly between points C and D, and when the bridge deck is cold, it occurs mainly between B and C. However, this difference is neglected in the load case formulation. The influence of the depth of the foundation (which affects the position of the level "E" in the figures) has also been investigated, as well as the effect of the tapering of the frame corner. Results from these comparisons are shown in Figures 8 and 9.

As the location of level E is varied in Figure 8, the figure shows five different lines for E, each corresponding to one analysis. Considering this difference in transition zone length, the analyses were considered to confirm the chosen division of structural parts. The variation of tapering shown in Figure 8 was considered to have a negligible influence on the temperature variation along the system line.

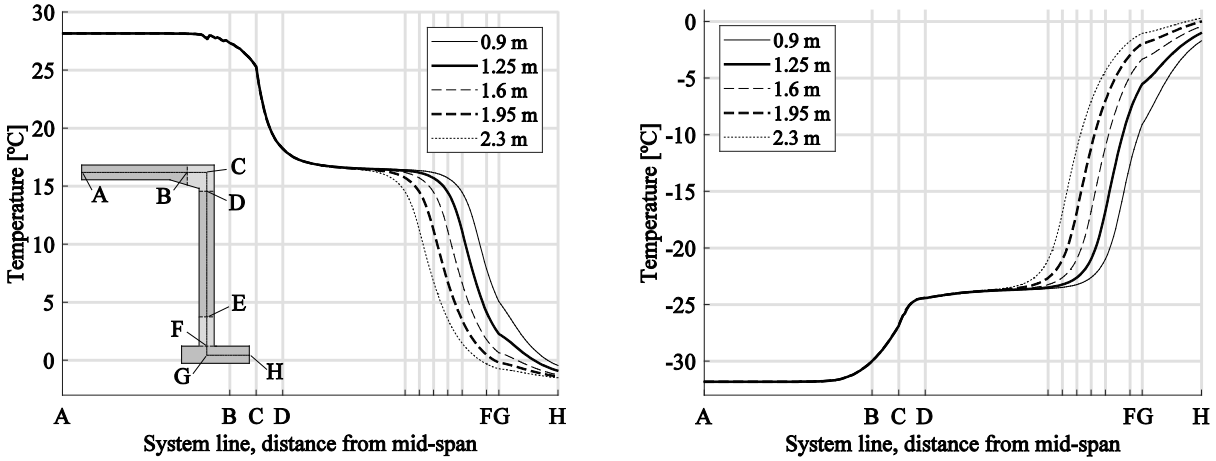


Figure 8. Temperature along the system line when the foundation depth was varied. The legends indicate the length of the transition zone between abutment and foundation, i.e. the distance between E and G.

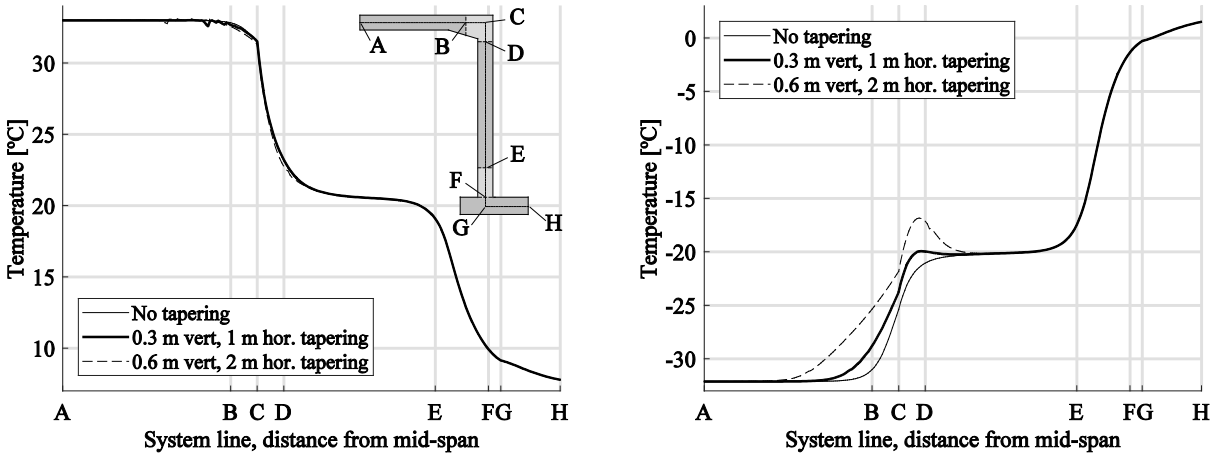


Figure 9. Temperature along the system line when the size of the tapering in the frame corner was varied. The tapering did not affect the position of the system line chosen in the illustration.

4. Crack widths due to restraint effects

This section presents findings regarding the widths of cracks for a quasi-permanent load combination, when cracking is caused by restraint effects. Only portal frame bridges subjected to self-weight, a difference in shrinkage between the foundations and the rest of the structure due to casting of the bridge in stages, and the general thermal load case presented in Section 2.2, were investigated. The shrinkage and thermal action are however the only loads which causes major tensile stresses in the transverse direction of the bridge. Horizontal soil pressure on the wing walls can also cause tensile stresses in the transverse direction, but the resulting stresses due to the at-rest earth pressure become relatively small. The earth pressure would however be larger in the case of a uniform temperature reduction, but this was not investigated in the present study.

The reason for investigating crack widths due to the thermal load case and differences in shrinkage is that the cracking response due to restraint effects is fundamentally different from cracking due to external loading. This is explained briefly in Section 4.2. The main conclusions of the study are summarized in Section 4.1.

4.1. Main conclusions

The main conclusions from the analyses of cracking in portal frame bridges are listed below. The listed conclusions are only valid for portal frame bridges subjected to differences in shrinkage between foundation and abutment due to casting of the bridge in stages, and the thermal load case presented in Section 2.

- Significant restraint effects only occur in the transverse direction. Therefore, the load cases only affect the reinforcement requirements in the transversal direction.
- Cracking in the bridge deck is unlikely to occur for the investigated load effects.
- Cracks might form in the top of the abutment. These cracks could become wider than the limit values given in [14], especially if the cross-sectional height of the abutment is smaller of that of the bridge deck.
- The largest cracks in the study were found in the bottom of the abutment, as the thermal action and shrinkage difference both causes tensile stresses to form in this area.
- Cracking in the foundation was not investigated, as the shape of the foundation can vary significantly, making it difficult to draw general conclusions. However, cracking will be less severe than in the lower part of the abutment, as a difference in shrinkage will cause compressive stresses in the foundation.
- The resulting crack widths were smaller than corresponding crack widths calculated without considering the reduction of stiffness due to cracking. The corresponding calculations were performed by running an FE-analysis with a linear elastic material model for concrete, and thereafter calculating crack widths using the section forces in the analysis in the expressions for crack width calculation in Section 7.3.4 of Eurocode 2-1-1 [11]. This is considered to be a common design approach today.
- The amount of reinforcement has a smaller impact on the crack widths in the investigated restraint cases than what is suggested in the equations in Section 7.3.4 of Eurocode 2-1-1 [11]. It could therefore be more effective to reconsider the geometry rather than to add extra reinforcement in bridges in which large restraint effects can be expected.
- In order to consider restraint effects more accurately in design, more research is needed. Two possible approaches to finding a design methodology can be suggested. The first approach is to find a mathematical expression between the results of non-linear FE-analyses and the results obtained by using linear elastic FE-analyses and calculating the crack widths according to Section 7.3.4 in Eurocode 2-1-1. The mathematical expression could then be used in

combination with the linear elastic design method and the same crack width calculations in order to account for the restraint effect. The second approach is to consider cracking due to the investigated restraint effects separately, and formulate a calculation methodology e.g. based on the theory for end restrained elements presented in Section 4.2 of this report.

4.2. Background – cracking due to restraint effects

Restraint stresses are stresses which appear when a structural element is prevented from obtaining its desired shape and volume. As shrinkage and thermal actions cause a desire to change the shape and volume, these actions will cause restraint stresses if the desired movements are prevented.

In order to illustrate the behavior of restrained structural parts, a reinforced concrete bar which is restrained from movements in both of its ends is studied when subjected to a decrease in temperature. If the bar is uncracked, the stiffness *EA* is constant along it, and the magnitude of the restraint force equals the restrained strain multiplied by the cross-sectional stiffness *EA*, as shown in Equation (1). In the equation, F is the restraint force, ε_R is the restrained strain, E is Young's modulus and A is the cross sectional area [15].

$$F = \varepsilon_R EA \tag{1}$$

However, when the reinforced concrete bar cracks, the stiffness is significantly reduced in the cracked sections, where the stiffness *EA* is only constituted by the reinforcement. Cracking will thereby reduce the stiffness of a structure and thus also reduce the restraint forces. However, Equation (1) can only be used directly to calculate restraint forces if ε_R, E and A are constant along the element, which is not the case if cracks have formed.

The effect of cracking on the restraint force in end-restrained bars is illustrated in Figure 10. In the figure, the end restrained reinforced bar is subjected to a decrease in temperature, ΔT, which causes restraint stresses. Due to force compatibility, the tensile force F must be constant along the entire bar. The bar can therefore be considered as a system of springs connected in a series, for which the total elongation corresponds to the restrained elongation. The force caused by the elongation thereby corresponds to the restraint force. Before cracking occurs, the stiffness k is equal in all the springs, assuming they are corresponding to parts of the reinforced bar with equal lengths. The strain ε is thereby also equal in the springs.

In the spring model in Figure 10, the first crack that forms causes a reduction of the stiffness of the spring in which the crack is situated. If the length of this spring corresponds to about half the total transmission length on both sides of the crack, the spring will now have a stiffness corresponding to the reinforcement bar only, while the stiffness of the other springs is unchanged. The formation of the crack does not affect the total elongation of the bar, as the total desired elongation due to the thermal action is unaffected by the cracking. However, the force in the element is reduced, as the stiffness of one of the springs is reduced. On the other hand, the strain, which was equal in the springs before cracking, is now increased in the weaker spring, and reduced in the other springs. This development is illustrated in Figure 10 (b). The model presented here is a simplified version of the model used in [16].

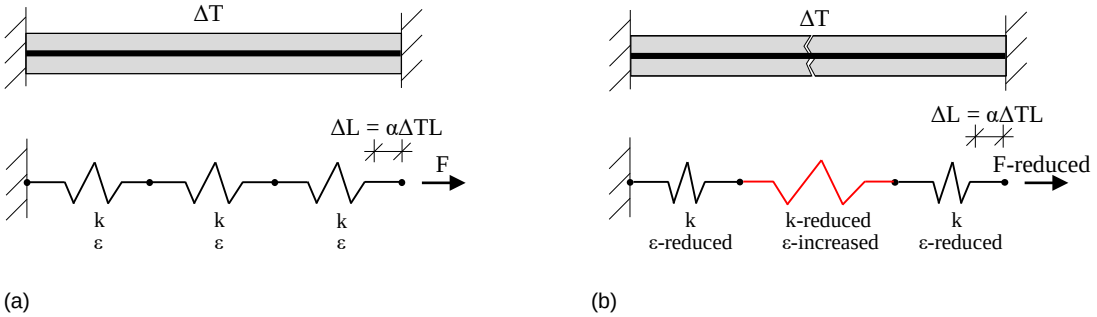


Figure 10. Illustration of a bar which is uncracked (a) or has one crack (b) with spring models of the bars shown below.

As restraint forces are reduced by cracking, the crack development phase will look different depending on whether restraint or non-restraint forces are acting on the structure. In order to illustrate this difference, another example of a reinforced concrete bar is shown, see Figure 11 (a). If the bar is subjected to a continuously increasing external load, the bar will go from uncracked to a stabilized crack pattern over a very small increase in loading, due to the small variations in concrete material properties along it, see Figure 11 (b). The elongation of the bar, ΔL, can be expressed as in Equation (2), which shows ΔL as a function of the force F, the length of the bar, L, and the average stiffness AE. As can be seen in the equation, the elongation is drastically increased for every new crack formed, as the stiffness of the bar is reduced.

$$\Delta L = FL/(AE) \tag{2}$$

If instead the elongation is continuously increased, the load – elongation relation will look different, as shown in Figure 11 (c). When the stiffness decreases in this case due to the formation of a crack, the force must also decrease in order for the elongation to remain unchanged. As the elongation continues to increase, a new crack will form each time the tensile strength of the concrete is reached.

The most important differences between the two cases illustrated in Figure 11 (b) and (c) are that in the restrained case, cracking reduces the stresses in the specimen, and that the development from one crack to the stabilized crack stage requires a significant increase in restraint. This difference in behavior is discussed in e.g. [15, 17-20]. The conclusion that a significant increase in restraint is required to reach the stabilized crack pattern after the first crack has formed means that it can normally not be assumed that a stabilized crack pattern is reached when restraint stresses are dominating. This implies that the stiffness, and therefore also the restraint stresses, are hard to predict in the specimen.

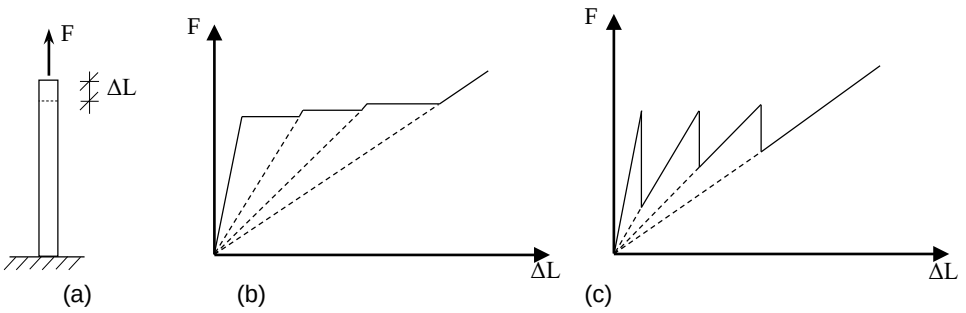


Figure 11. Comparison of load-deformation diagram of a reinforced concrete bar (a) subjected to a continuously increasing external load (b) or a continuously increasing elongation (c), corresponding to increasing restrained deformations. Figure after [15, 17-20].

Figure 12 shows the same load-elongation pattern as Figure 11 (c), but with the stiffness of the uncracked stage (stage I) and the stabilized crack stage (stage II) marked. The stage I stiffness is denoted EA_I, and the stage II stiffness is denoted EA_{II}. Figure 12 illustrates that if the stabilized crack pattern is not reached, the entire crack development process must be known in order to estimate the number of cracks and thereby the stiffness and stresses in the model. Using the stage I stiffness would in this case overestimate the stresses, while using the stage II stiffness would underestimate the stresses. This is illustrated by the forces F_I and F_{II} in Figure 12, which are estimations of the restraint force F_R at the restrained elongation ΔL_R, using stage I and II stiffnesses, respectively. The stage II stiffness can be estimated beforehand by considering the distance between cracks at the stabilized cracking stage. As the expressions for crack width in Section 7.3.4 of Eurocode 2-1-1 do not describe the reduction of restraint stresses due to cracking, there is no consideration of this stress reduction in the common design method. This implies that if the effects of cracking are not considered in another way during design, the resulting stresses from the calculations will correspond to F_I in Figure 12, and thus risk being significantly exaggerated.

For structures such as base restrained walls and portal frame bridges, the development of cracking due to restraint effects becomes more complicated, as the desired movements are prevented to different

extents in different parts of the structures. This is described by the degree of restraint, *R*, which is calculated as the restrained elongation divided by the total desired elongation. In a point where deformations are completely restrained, the degree of restraint thus equals one, and in a point where the structure can deform freely, the degree of restraint equals zero. The fundamental variation of degree of restraint in a base restrained wall and a portal frame bridge is illustrated in Figure 13.

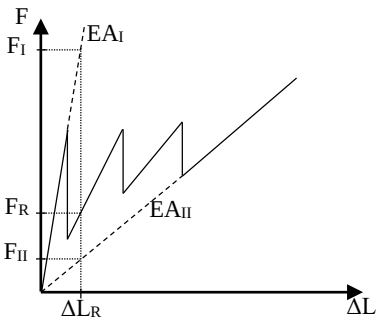


Figure 12. Load-deformation diagram of an end-restrained bar subjected to restraint effects. A restrained elongation ΔLR causes the restraint force FR. Estimating FR by using the state I stiffness EAI or by using state II stiffness EAII gives the inaccurate results FI or FII respectively.

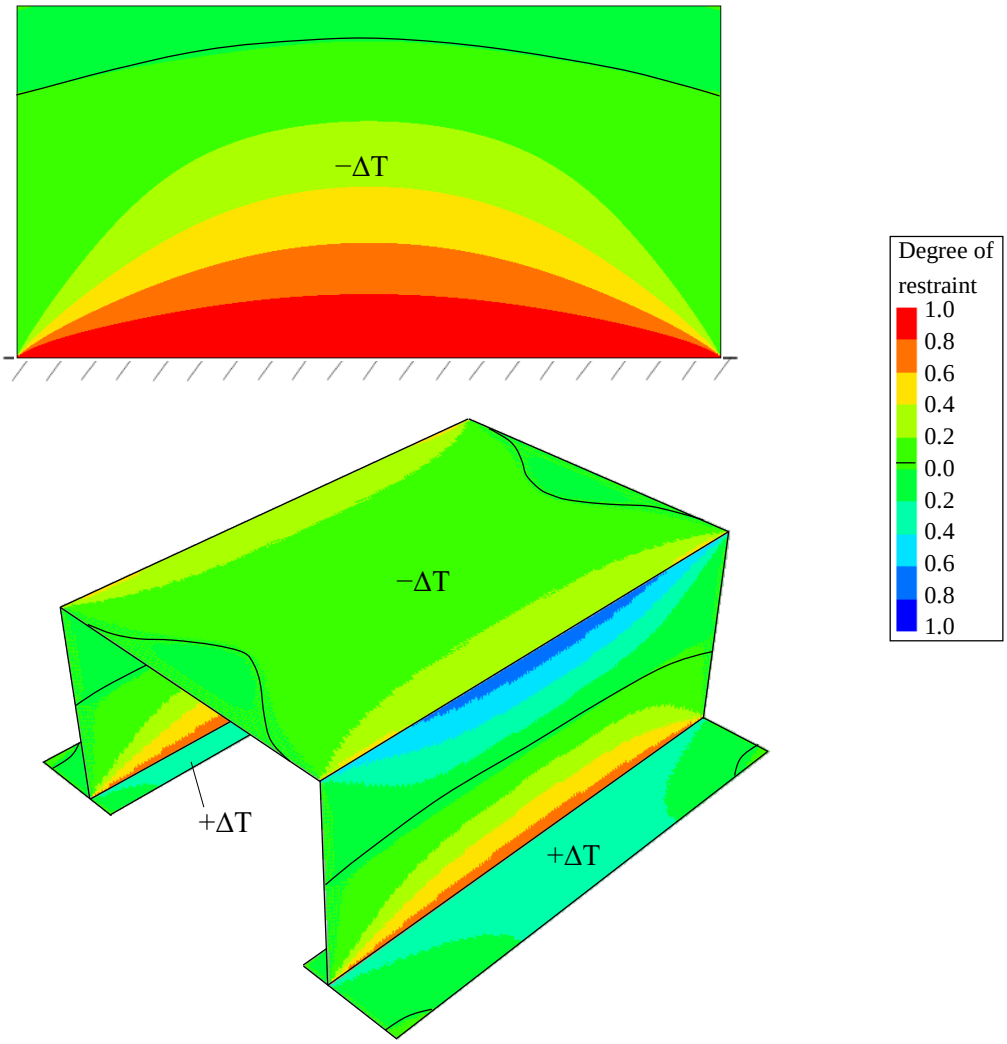


Figure 13. Variation of degree of restraint in a base restrained wall with a fully restrained bottom edge (top) and a portal frame bridge (bottom) in the width direction (transversal direction, see Figure 1) of the structures.

4.2.1. Influence of creep

Creep describes the increase in deformations that occur over time when structures are subjected to long-term loading. If a structure is subjected to restraint effects, creep will instead lead to a gradual decrease of stresses in the structure. In design calculations, creep effects are often considered as a reduction of Young's modulus, as shown in Equation (3), where E_c is the initial Young's modulus, ϕ is the creep coefficient and $E_{c,eff}$ is Young's modulus with consideration taken to creep effects. The development of creep over time is thereby disregarded in the calculations, and the final deformation (or the stress value, in restraint situations) is instead determined directly from the load effect (or the imposed strain, in the case of restraint) [21].

$$E_{c,eff} = \frac{E_c}{1+\phi} \tag{3}$$

Considering the final creep value this way in serviceability limit state often corresponds to a worst case scenario in non-restraint cases, as it leads to larger deformations in the structure. However, this is not the case regarding restraint effects.

As cracking occurs for a certain stress value in the concrete, the assumption of the final creep value having developed means that a larger restrained strain is required in order to cause cracking. If this is the case, the larger strain will cause a larger stress in the reinforcement before cracking occurs, which means that the cross-sectional force required to cause cracking increases. This in turn leads to an increase of crack widths. This is illustrated in Figure 14, which shows the restraint force and crack widths in a reinforced concrete bar subjected to tension for different creep coefficients. The bar is made of concrete class C30, has a length of 2 m and a cross-sectional area of 0.1x0.1 m^2, and is reinforced with one reinforcement bar with a diameter of 16 mm.

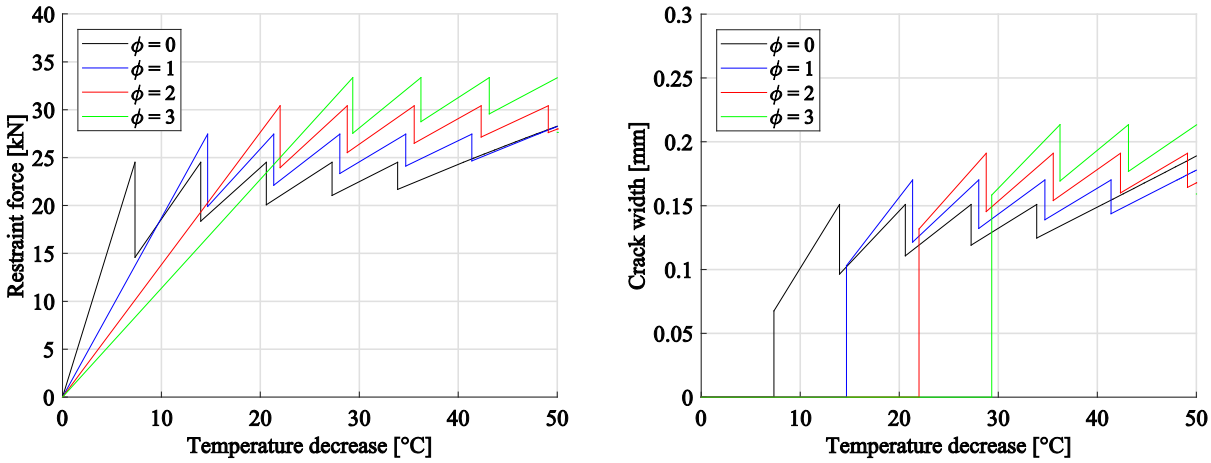


Figure 14. Diagram showing the restraint force (a) and the crack width (b) ina reinforced concrete bar as a function of the temperature decrease, for four different values of creep.

However, in linear elastic design models, the reinforcement is generally omitted, as the required amount of reinforcement is to be decided. Omitting the reinforcement means that the increase in force and thus crack widths is not captured when creep is considered as a reduction of Young's modulus according to Equation (3). Such an analysis can thus capture the restraint effect required for cracking correctly, assuming the creep coefficient is correct and cracking does not occur before the creep has developed, but will underestimate the resulting crack widths. Also, subsequent cracking is delayed, as the reduction of the restraint force due to cracking is overestimated. This is illustrated in Figure 15, which shows the restraint force and crack widths for the cases with $\phi=0$, $\phi=1$ and estimating $\phi=1$ by halving the load effect in the model. The reduction of the load effect is in this case achieved by halving the coefficient of thermal expansion. This reduction of the load effect corresponds to halving the Young's modulus (i.e. assigning $\phi=1$) in a linear elastic model which does not include reinforcement.

The effect of creep on reinforced concrete structures subjected to restraint effects is thus that cracking is delayed as the reduction of the Young's modulus of concrete reduces the overall stiffness of the structure. However, if cracking occurs, fewer cracks will form, and the structure will experience a larger influence of the reinforcement, which leads to larger crack widths. This effect is not captured in the common design method, in which the reinforcement is not included in the FE-analysis. Also, assuming that the final creep value have developed before cracking occurs is on the unsafe side, thus possibly overestimating the restraint effect required for cracking.

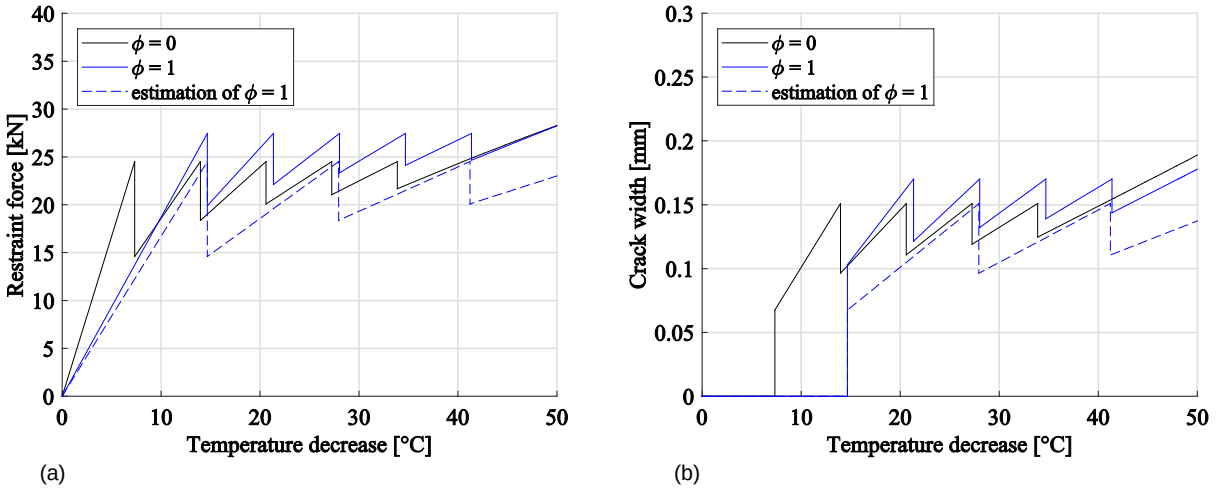


Figure 15. Diagram showing the restraint force (a) and the crack width (b) in a reinforced concrete bar for different values of creep and different methods of considering creep in the model.

4.3. Methodology – FE analyses

Cracking due to restraint effects was investigated within the research project mainly by preforming FE-analyses with a non-linear material model for concrete in tension, i.e. cracks could form in the analyses. The reinforcement was modeled as discrete bars, and a bond-slip relationship was used for concrete – reinforcement interaction. At first, cracking in base restrained walls was analyzed, and compared with full scale tests performed in [22, 23]. The results of the comparison are presented in [5], along with a more detailed description of the analysis model. Plane stress elements were used in the walls, and the foundations of the walls were also included in the structural models. Interface elements were placed between the walls and foundations, which described the stress-slip relation both along the edge and perpendicular to the edge in the vertical direction. The shear stiffness of the edge was based on expressions given in [24], while the vertical stiffness was based on the stiffness of 0.2 m long parts of the reinforcement bars connecting the structural parts. Interface elements were also used below the foundation, to describe the vertical deformation of the soil due to the self-weight combined with the curving of the foundations, which was caused by the thermal action and shrinkage acting on the walls. The interface elements had a vertical stiffness of 50 MN/m^3 for downwards movements, while the stiffness was zero for movements upwards, as well as for horizontal expansion and contraction.

After the verification of the model, the analysis model was developed further into a model of a portal frame bridge. In this model, the plane stress elements were replaced with curved shell elements. The loads included in the model were self-weight, a difference in shrinkage between foundation and abutment of 10^{-4}, corresponding to the foundations being cast about 250 days prior to the casting of the rest of the bridge, and the general thermal load case presented in Figure 2. The thermal action was applied last, by adding the characteristic values in order to induce cracking, and thereafter reducing the load to its quasi-permanent value before crack widths were measured. Horizontal earth pressure on e.g. the wing walls could cause tensile stresses in the transverse direction of the abutment, but this earth

pressure was found to be small for the given thermal load case. The earth pressure would be larger when the bridge is subjected to a uniform reduction of temperature, which is not considered in this study. Therefore, horizontal earth pressure was omitted from the analyses. Other actions, which do not cause stresses in the transverse direction, were also omitted from the study. The self-weight and shrinkage were considered as permanent loads, i.e. they were applied before the thermal action and were not adjusted. Although quasi-permanent values were used, creep was not included in the analyses. The influence of creep is discussed in Section 4.2.1.

4.3.1. Analysis results, cracking in the corner region

FE-analyses using a linear-elastic material model for concrete showed that transversal tensile stresses in the bridge decks due to the investigated load cases can reach 1.75 MPa for the characteristic temperature difference, see Figure 16, but are unlikely to become much larger. The value was obtained for a bridge with a large width in relation to its height, as the bridge deck was 8 m in longitudinal direction, the height of the abutments was 6 m and the width of the bridge was 20 m. It should also be noted that the result was obtained when both the shrinkage difference and the temperature difference between abutment and foundation were neglected, as these load effects cause compressive stresses in the transversal direction in the bridge deck. The maximum tensile stress in the bridge deck did not increase significantly when the width of the bridge was doubled, for which reason the obtained stress is assumed to show that the tensile stress in the bridge deck will not supersede the characteristic tensile strength of the concrete, which is e.g. 2.5 MPa for concrete quality C40. In the analysis, the cross-sectional heights of bridge deck and abutment were the same. In real cases, the cross-sectional height of the bridge deck is usually either equal to that of the abutment or larger. In the case that the cross sectional height of the bridge deck is larger, the degree of restraint will decrease in the bridge deck and the resulting stresses will thus also decrease.

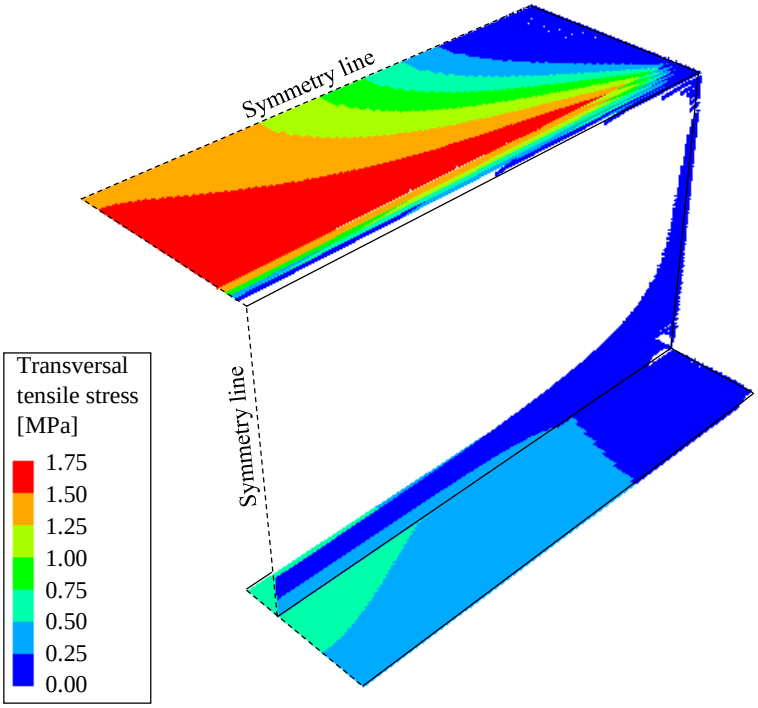


Figure 16. Transverse tensile stresses at the middle of the cross-sectional height of a portal frame bridge subjected to the characteristic temperature difference between bridge deck and abutment shown in the bottom of Figure 2.

The tensile stresses resulting from the general load case become larger in the top of the abutment than in the bridge deck. The reason for this is the way the structure deforms, and relates to the symmetry condition in the middle of the bridge span. Also, the tensile stresses in the top of the abutment are increased by the temperature difference between the abutment and the foundation.

The maximum tensile stress in the abutment obtained in the model was 3.0 MPa, which is lower than the mean tensile strength but higher than the characteristic tensile strength of e.g. concrete quality C40. For this reason, cracking was investigated using non-linear FE-modeling. However, as non-linear FE-modeling is generally performed using mean values of material parameters in order to obtain the most accurate structural response, and as the mean strength of concrete C40 used in the analyses, 3.5 MPa, is higher than the actual load effect in this case, the load was multiplied with a factor 1.4, corresponding to the difference between mean and characteristic tensile concrete strength. This way, the structure will crack, and the load effects are unlikely to be underestimated.

Figure 17 shows cracks in the top of the abutment in two walls with different widths. Minimum reinforcement according to Equation (7.1) in Eurocode 2-1-1 [11] with *k* calculated according to [14] was placed in the abutments, corresponding to φ16 cc 195 mm both horizontally and vertically and on both sides of the cross sectional system line. In the analysis of the 10 m wide bridge, only one crack appeared, which formed beside the symmetry line and reached a maximum width of 0.10 mm for the quasi-permanent load combination. The fracture energy of this row of elements was halved in order to account for the equal crack forming on the other side of the symmetry line. By halving the fracture energy, it was considered that the entire crack width was captured in the model. When the bridge was 20 m wide, two cracks formed, one beside the symmetry line and the other about 3.7 m away from it. The maximum crack width was in this case 0.12 mm.

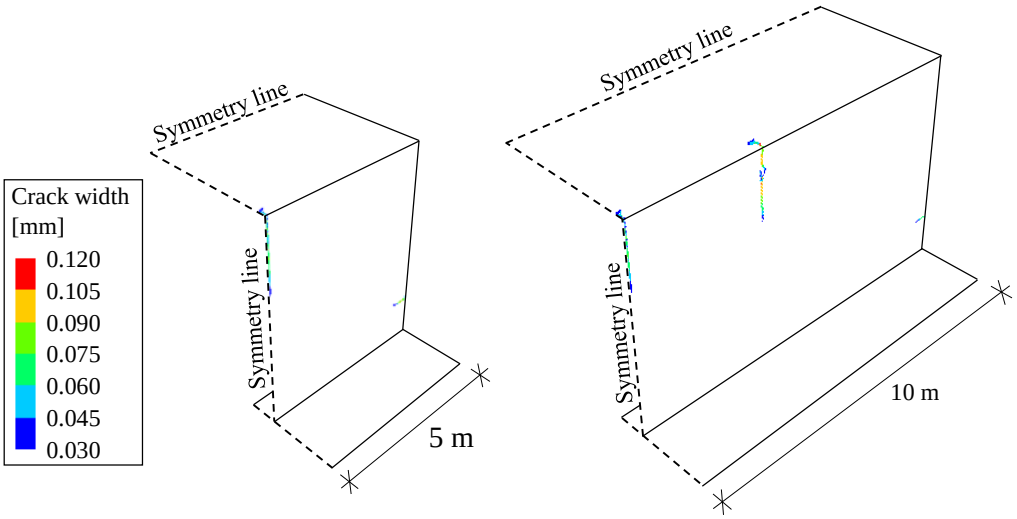


Figure 17. Crack patterns and crack widths when cracking in the top edge of the abutment was investigated. Results are shown for the quasi-permanent load combination, after the thermal load has been reduced from its characteristic value.

The characteristic crack width, which shall be compared with the demands on crack width limitation, was calculated as the mean crack width for the quasi-permanent temperature difference multiplied with 1.3, according to [25]. In this case, the largest crack width in each wall multiplied with 1.3 renders crack widths of 0.13 mm and 0.16 mm, respectively. Choosing the maximum crack width from the analyses as the mean crack width to be multiplied with 1.3 means that a safety margin is added to the results, which thereby are conservative estimates. As crack width limitations are often 0.15 mm or 0.20 mm [14], it can be concluded that the load case combined with minimum reinforcement can lead to crack widths close to the limit of what is considered as being acceptable.

It should be noted that if the bridge deck and abutment have different cross-sectional height, the part with smaller cross-sectional height experiences a larger degree of restraint. As abutments are in some cases made with smaller cross-sectional heights than bridge decks, especially in bridges with long spans, this likely increases the crack widths in the abutments. Also, for the investigated cases, the stress variations over the cross-sectional height were small, and the cracks forming in the analysis model were through cracks. The use of curved shell elements means that stress transfer in the direction of the cross-sectional height is not modeled. This means that in the model, the entire cross-sectional height is

considered as the effective cross-sectional height. For the specific structural model investigated in this study, this had only a small impact on the result, as the cross-sectional height was 400 mm and the transverse reinforcement cover was 72 mm on both sides.

In conclusion, it is likely that the bridge decks of portal frame bridges will not crack due to differences in temperature between the structural parts. The top part of the abutments may however do so, and the cracks might become wider than allowed, especially if the abutment cross-sectional height is smaller than that of the bridge deck.

4.3.2. Analysis results, cracking in the lower part of the abutment

The thermal action presented in Figure 2 (for the case when the foundation is warmer than the abutment) interacts with the difference in shrinkage in causing tensile stresses in the lower part of the abutment. Therefore, the largest cracks in the analyses were found in this area. The analyses made were focusing on investigating the effect of the reinforcement amount. Seven analyses were made using different cc-distances for the reinforcement in both vertical and horizontal direction. Reinforcement with 16 mm diameter was used also in this case, and placed on both sides of the system line. The resulting maximum characteristic crack widths are shown in Figure 18, as a function of the cc-distance of the reinforcement. In a corresponding linear elastic analysis using quasi-permanent load values, the largest transverse tensile stress value was 2.7 MPa after being smoothed out according to [26]. The crack width calculated from this result using equations in section 7.3.4 in Eurocode 2-1-1 [11] is also shown in the figure as a function of the cc-distance. As can be seen in the figure, the difference in crack width between the results obtained using the conventional design method and the non-linear analyses increases with reduced reinforcement ratio.

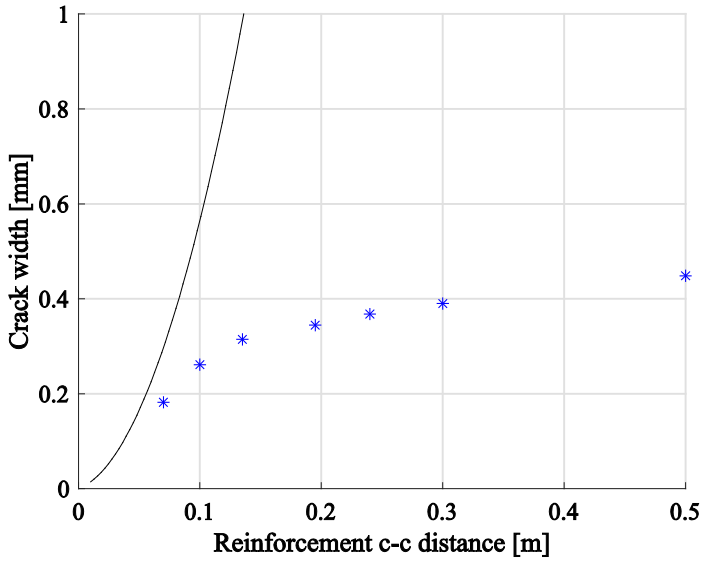


Figure 18. Characteristic crack widths according to non-linear analyses (stars) and crack widths calculated using Eurocode 2-1-1 [11] and stresses from linear elastic analyses (line).

Not only is the crack widths considered unacceptable when the minimum reinforcement amount is used. In fact, all the obtained maximum characteristic crack widths are larger than the strictest demand on crack widths in Swedish bridges, i.e. 0.15 mm [14]. Therefore, cracks which are unacceptably wide could form in Swedish portal frame bridges due to the investigated restraint effects.

Figure 18 shows that the reinforcement amount does not influence the crack widths in restraint situations as strongly as in non-restraint situations. One reason for this is that when the reinforcement ratio is increased, the stiffness is also increased, which in turn increases the restraint and also the crack widths. Another possible explanation is that in this type of restraint situation, the restraining edge could be significantly limiting the widths of the cracks. This is suspected as the cracks cannot propagate over the edge between structural parts, due to the structural part on the other side of the edge being in

compression. Another reason for the increasing difference with decreasing reinforcement amount is that in the non-linear analyses, the total restrained deformation in the abutment is considered. This deformation can be estimated to 0.68 mm as is shown in Equation (4). In the equation, ΔL_R is the total restrained deformation, ΔT corresponds to the combined difference in desired elongation between abutment and foundation due to thermal action and shrinkage, α is the coefficient of thermal expansion, R is the average degree of restraint at the level of height in the abutment where the restraint is largest, and L is the length of the bridge in the transversal direction. As the crack width should not be larger than the total restrained deformation, it is not surprising that the crack widths in Figure 18 according to the non-linear analyses remain below 0.68 mm. Further, as there are more than one crack in the walls (see Figure 19) which together should not exceed this value, it is even less surprising.

$$\Delta L_R = \Delta T \alpha R L \approx (10 + 7) \cdot 10^{-5} \cdot 0.4 \cdot 10 = 0.68 \cdot 10^{-3} m \quad (4)$$

The small influence of the reinforcement amount can also be seen by observing the crack pattern in the abutments, which are relatively similar. In the seven analyses, three cracks formed in the abutments in all cases except for when the cc-distance was 70 mm, in which case five cracks formed. The crack patters for the cases with φ16 cc 500 mm, φ16 cc 195 mm, φ16 cc 100 mm and 70 mm are shown in Figure 19. The smaller influence of reinforcement amount on crack widths was also seen in the tests performed by [22, 23], as is also shown in [1]. Note that the crack widths shown in Figure 19 are the actual results of the analyses, while the results in Figure 18 shows the characteristic crack widths, which in this case have been calculated as the maximum crack widths in the analyses multiplied with 1.3 [25].

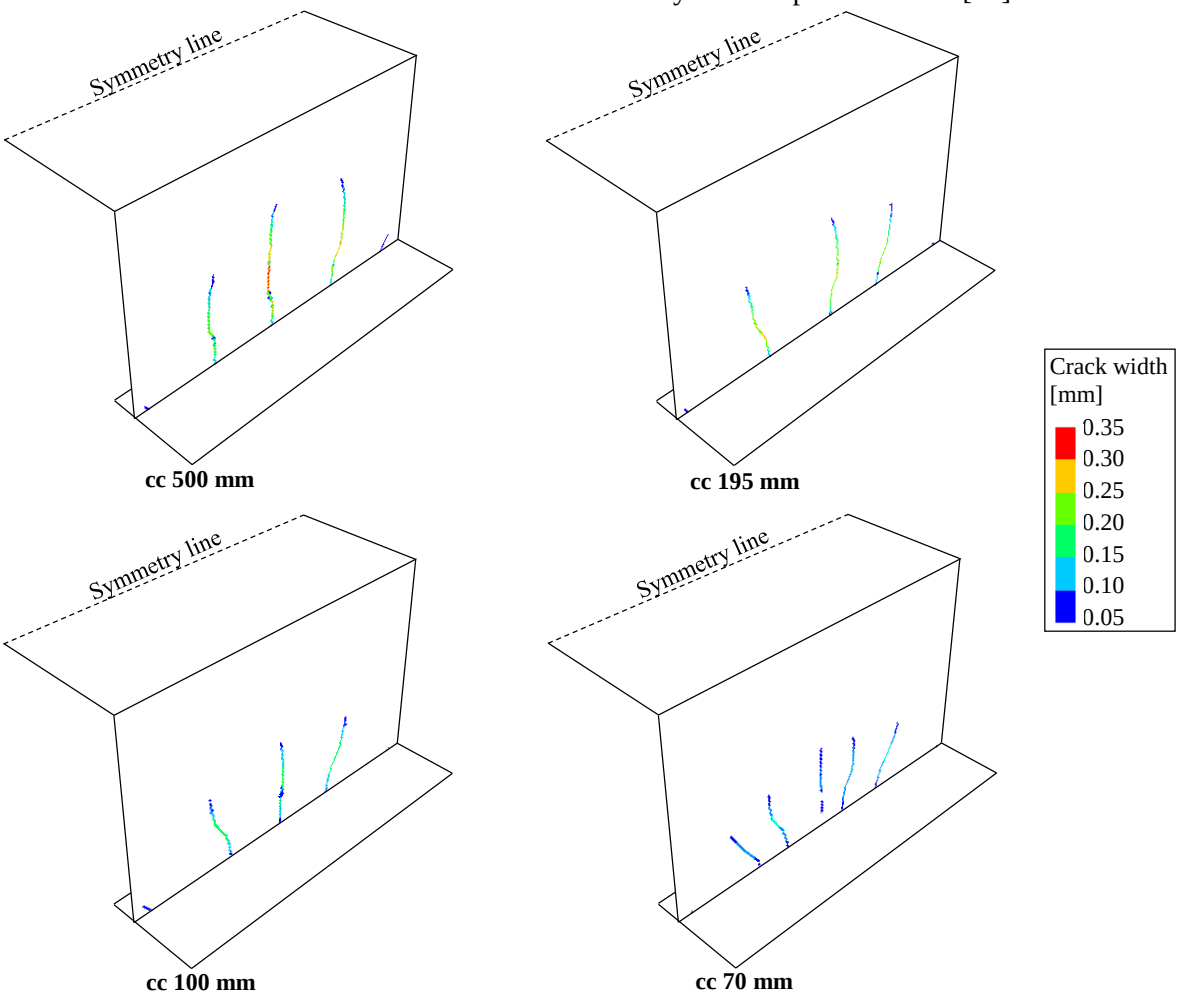


Figure 19. Crack pattern in portal frame bridge with reinforcement φ16 with the cc-distances 500 mm (top left), 195 mm (top right), 100 mm (bottom left) and 70 mm (bottom right) when subjected to a shrinkage difference, and having been subjected to characteristic loads which thereafter were reduced to quasi-permanent load values. The given crack widths have not been recalculated to characteristic values.

When the same minimum reinforcement is used as in Section 4.3.1 (φ16 cc 195 mm on both sides of the cross section) the characteristic crack width (0.34 mm) is larger than the limit values for crack widths in Swedish bridges (often 0.15 or 0.20 mm [14]). It should however be noted that the crack width resulting from the analysis is dependent on the bridge geometry, i.e. no general conclusion could be drawn from the results. Also, along edges where differences in shrinkage due to casting of the bridge in stages can appear, extra reinforcement of 5 φ16 bars are generally placed in Sweden on both sides of the cross section in the newer part, with a 200 mm cc-distance. This means that in a real case, more reinforcement should be placed in the lower part of the abutment. However, this reinforcement is placed lower than the height levels of the largest crack widths in the model, and when the reinforcement is included, the maximum characteristic crack width is only reduced to 0.32 mm.

Figure 20 shows the maximum characteristic crack widths in different steps of the analyses. The graph shows how the maximum crack width develops under loading up to characteristic load and unloading to the quasi-permanent load level. The difference in strain between abutment and foundation, shown on the horizontal axis in the figure, is caused by both shrinkage and thermal action. It can be seen in Figure 20 that the crack widths did not increase linearly with the increasing restrained strain, as is suggested in e.g. Eurocode 2-3 [27]. The reason for the crack widths not being proportional to the applied restraint loads is likely the reduction of restraint that occurs as the structure cracks. The effect of reducing the thermal action from its characteristic value to the quasi-permanent value is also visible in the figure.

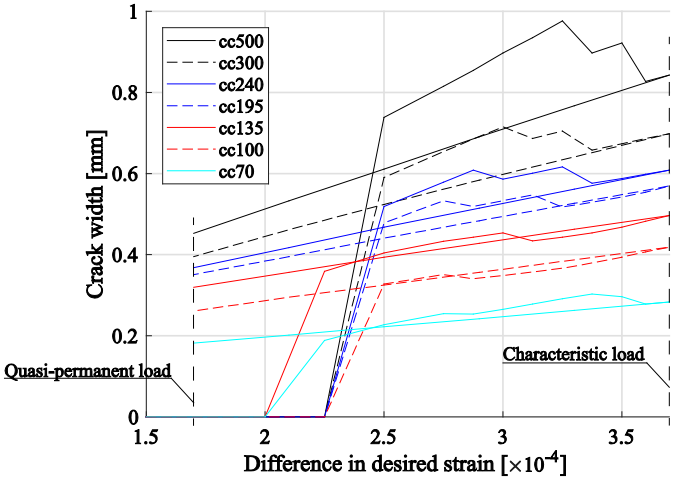


Figure 20. Development of crack widths in analyses of cracking in the lower part of the abutment in portal frame bridges. The horizontal axis shows the corresponding temperature difference between abutment and foundation, i.e. the combined effect of shrinkage and thermal action.

It should be noted that in many of the analyses, the largest crack width appeared before the largest restraint load was applied. This indicates that the crack width for quasi-permanent action would have been larger, if the load had not been increased all the way to the characteristic value prior to the load reduction. In the case of cc 500 mm in Figure 20, the largest quasi-permanent crack width would have been obtained if the load increase was stopped before the formation of the second crack, at a desired strain difference of $3.25 \cdot 10^{-4}$. This can also be understood as since the formation of additional cracks reduces the stiffness of the structure, the largest stiffness of a cracked structure is obtained when only one crack is present.

An investigation of the effect of changing the width of the bridge was also made. The analysis model with reinforcement cc 300 mm was used, and the width of the bridge was changed from 10 m to 5 m and 15 m, respectively. A comparison of the results from using both non-linear analysis and the conventional design method is shown in Figure 21. The crack widths in the figure are not recalculated into characteristic values. The maximum characteristic crack widths were 0.25 mm, 0.39 mm and 0.51 mm respectively in the analyses with widths of 5 m, 10, and 15 m respectively. The comparison shows that increasing the width of the bridge increased the crack widths significantly, but the results are likely

sensitive to the exact number of cracks forming in the analyses, and can be different for different bridge geometries.

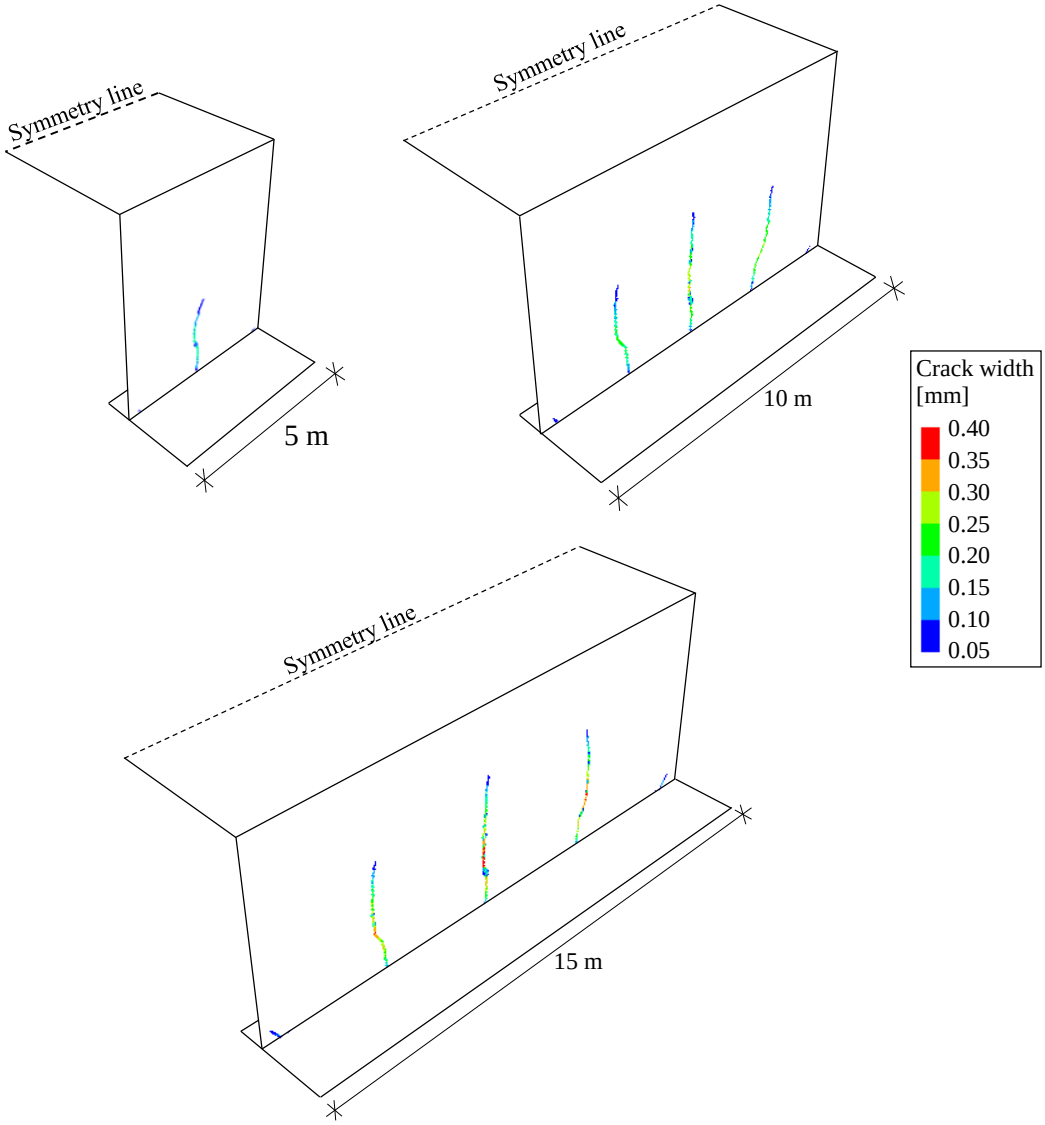


Figure 21. Illustration of cracks in portal frame bridges width different widths.

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